Laplace Transforms

Turning the Calculus of Differential Equations into Algebra

M3503 Ordinary Differential Equations • Unit 3
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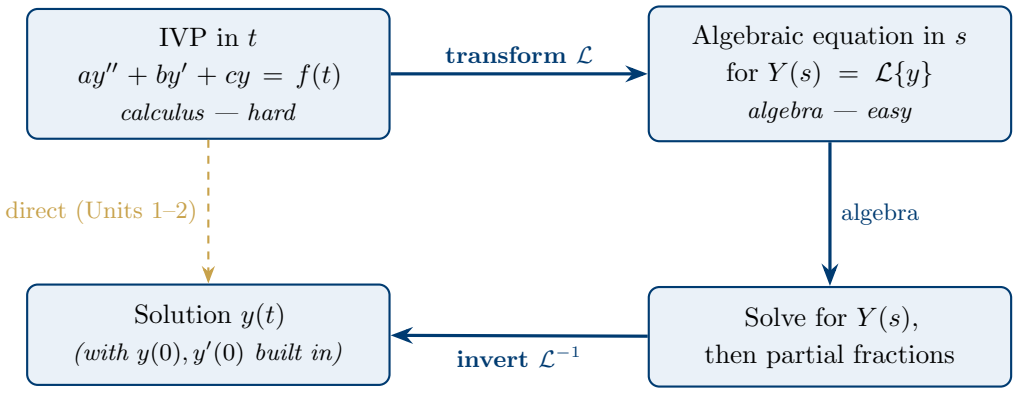
1. Why transform at all?

The methods of Units 1 and 2 — integrating factors, characteristic equations, undetermined coefficients, inverse operators — solve large classes of linear ODEs. But three situations strain them:

- **Initial-value problems.** We first find a general solution, then solve extra equations to pin the arbitrary constants to the data $y(0), y'(0)$. The bookkeeping grows with the order.
- **Forcing that switches or spikes.** A voltage switched on at $t = a$, or a sharp hammer-blow, is piecewise or singular; feeding it into undetermined coefficients is awkward at best.
- **Systems.** Several coupled ODEs must be untangled by hand.

The *Laplace transform* handles all three by a single strategic move: it converts the **calculus** of a differential equation into **algebra**. You transform the problem into a new variable s, where differentiation becomes multiplication and the ODE becomes an ordinary algebraic equation; you solve that by algebra; then you transform back.

The right analogy: logarithms. Before calculators, a hard multiplication $x \times y$ was done by mapping to "log-world" ($\log x + \log y$, an *easy* addition) and mapping back with an antilog. The Laplace transform plays the same trick for differential equations: map to the "s-domain", where the hard operation (differentiation) becomes an easy one (multiplication by s), solve, and map back.



The dashed route is the direct attack; the solid route goes "the long way round" — but every leg of it is easy, and the initial data enter for free.

2. The definition

Laplace transform

For a function $f(t)$ defined on $t \geq 0$, its Laplace transform is the function of a new variable s,

$$\mathcal{L}\{f(t)\}(s) \;=\; F(s) \;=\; \int_0^\infty \mathrm{e}^{-st}\, f(t)\,\mathrm{d}t,$$

defined for all s large enough that the integral converges.

Three things to notice. The transform *integrates* t *away*: the output depends only on s. The lower limit 0 means Laplace looks only at $t \geq 0$ — it is built for initial-value problems posed at $t = 0$. And the kernel e^{-st} is a decaying weight that tames growth in f, provided s is large enough.

When does it exist? It is enough that f be piecewise continuous on $[0, \infty)$ and of *exponential order* — that is, $|f(t)| \leq M\mathrm{e}^{ct}$ for some constants M, c and all large t. Then $F(s)$ exists for every $s > c$. This covers every function met in this course: polynomials, exponentials, sines and cosines, and their products and sums. One consequence worth remembering: $F(s) \to 0$ as $s \to \infty$, so an expression like $\frac{s}{s+1}$ can never be a Laplace transform.

3. Transforms straight from the definition

The whole method rests on a short table, and the table is built by evaluating the integral once for each basic function.

The constant 1 and the exponential e^{at}

$$\mathcal{L}\{1\} = \int_0^\infty \mathrm{e}^{-st}\,\mathrm{d}t = \left[-\tfrac{1}{s}\mathrm{e}^{-st}\right]_0^\infty = \frac{1}{s} \quad (s > 0).$$

$$\mathcal{L}\{\mathrm{e}^{at}\} = \int_0^\infty \mathrm{e}^{-(s-a)t}\,\mathrm{d}t = \frac{1}{s-a} \quad (s > a).$$

Powers t^n

Integration by parts gives $\mathcal{L}\{t\} = \int_0^\infty t\,\mathrm{e}^{-st}\,\mathrm{d}t = \left[-\tfrac{t}{s}\mathrm{e}^{-st}\right]_0^\infty + \tfrac{1}{s}\int_0^\infty \mathrm{e}^{-st}\,\mathrm{d}t = \tfrac{1}{s^2}$. Repeating the parts step n times yields the general rule

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{\,n+1}} \qquad (n = 0, 1, 2, \ldots).$$

Sine and cosine — via one complex transform

From $\mathcal{L}\{\mathrm{e}^{at}\} = \frac{1}{s-a}$ with $a = i\omega$,

$$\mathcal{L}\{\mathrm{e}^{i\omega t}\} = \frac{1}{s - i\omega} = \frac{s + i\omega}{s^2 + \omega^2}.$$

Taking real and imaginary parts (using $\mathrm{e}^{i\omega t} = \cos\omega t + i\sin\omega t$ and linearity),

$$\mathcal{L}\{\cos\omega t\} = \frac{s}{s^2 + \omega^2}, \qquad \mathcal{L}\{\sin\omega t\} = \frac{\omega}{s^2 + \omega^2}.$$

4. The two properties that do the work

Linearity

The integral is linear, so the transform is too:

$$\mathcal{L}\{\alpha f(t) + \beta g(t)\} = \alpha F(s) + \beta G(s).$$

This is why a table of a handful of functions suffices: any linear combination transforms term by term.

The derivative rule — the engine of the method

Integrating $\int_0^\infty \mathrm{e}^{-st} f'(t)\,\mathrm{d}t$ by parts, and using $F(s) \to 0$ decay to kill the boundary term at infinity, gives the property that makes everything work:

Transform of a derivative

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0),$$
$$\mathcal{L}\{f''(t)\} = s^2F(s) - sf(0) - f'(0),$$

and in general $\mathcal{L}\{f^{(n)}\} = s^nF(s) - s^{n-1}f(0) - \cdots - f^{(n-1)}(0)$.

Read this carefully: differentiating in t becomes *multiplying by* s in the s-domain, and the initial values $f(0), f'(0), \ldots$ drop in automatically. A differential equation therefore turns into an algebraic one in $Y(s)$ that already knows the initial data — no arbitrary constants to chase at the end.

5. Coming back: inverse transforms and partial fractions

Solving in the s-domain leaves us holding $Y(s)$; the last leg is the **inverse transform** $y(t) = \mathcal{L}^{-1}\{Y(s)\}$. In practice this is: *split $Y(s)$ by partial fractions until every piece matches a table entry, then read the table backwards.* Three techniques do almost all of this work; each is stated, justified, and drilled below — and, so you can see where it turns up, each example carries a short IVP that produces exactly that inversion.

The first shifting theorem (the s-shift)

First shifting theorem

$$\mathcal{L}\{\mathrm{e}^{at} f(t)\} = F(s-a), \qquad \Longleftrightarrow \qquad \mathcal{L}^{-1}\{F(s-a)\} = \mathrm{e}^{at}\, f(t).$$

Multiplying by e^{at} in the t-domain *shifts* the transform by a in the s-domain.

Why it holds. Fold the exponential into the kernel of the defining integral:

$$\mathcal{L}\{\mathrm{e}^{at} f(t)\} = \int_0^\infty \mathrm{e}^{-st}\,\mathrm{e}^{at} f(t)\,\mathrm{d}t = \int_0^\infty \mathrm{e}^{-(s-a)t} f(t)\,\mathrm{d}t = F(s-a).$$

How to use it — both directions. *Forward:* to transform $\mathrm{e}^{at} \times$ (something familiar), transform the "something" to get $F(s)$, then replace every s by $s-a$. *Inverse (the workhorse):* whenever a transform's s appears *only* in the combination $s-a$ — a denominator like $(s-a)^n$, or a completed square $(s-a)^2 + b^2$ — peel off the shift, invert the un-shifted $F(s)$, and multiply the result by e^{at}.

Example 1 — forward

$\mathcal{L}\{e^{3t}t^2\}$.

Start from $\mathcal{L}\{t^2\} = \dfrac{2}{s^3}$. The factor e^{3t} shifts $s \mapsto s-3$:

$$\mathcal{L}\{e^{3t}t^2\} = \frac{2}{(s-3)^3}.$$

Applied in an IVP. Read backwards, this inverts a triple real pole. Solve $y''-6y'+9y = e^{3t}$, $y(0)=y'(0)=0$ (forcing at the double root 3 — resonance): transforming gives $(s-3)^2Y = \frac{1}{s-3}$, so $Y = \frac{1}{(s-3)^3}$ and $y(t) = \frac{1}{2}t^2e^{3t}$.

Example 2 — forward

$\mathcal{L}\{e^{-2t}\cos 4t\}$.

Here $\mathcal{L}\{\cos 4t\} = \dfrac{s}{s^2+16}$, and e^{-2t} shifts $s \mapsto s+2$ (that is, $a=-2$):

$$\mathcal{L}\{e^{-2t}\cos 4t\} = \frac{s+2}{(s+2)^2+16}.$$

Applied in an IVP. The same shifted form, inverted, solves $y''+4y'+20y = 0$, $y(0)=1$, $y'(0)=-2$ (roots $-2\pm 4i$): here $(s^2+4s+20)Y = s+2$, so $Y = \frac{s+2}{(s+2)^2+16}$ and $y(t) = e^{-2t}\cos 4t$.

Example 3 — inverse

$\mathcal{L}^{-1}\left\{\dfrac{6}{(s+1)^4}\right\}$.

The s appears only as $s+1$, so this is $F(s+1)$ with $F(s) = \dfrac{6}{s^4}$. Since $\mathcal{L}^{-1}\left\{\dfrac{6}{s^4}\right\} = t^3$ (because $\mathcal{L}\{t^3\} = 6/s^4$), the shift attaches e^{-t}:

$$\mathcal{L}^{-1}\left\{\frac{6}{(s+1)^4}\right\} = e^{-t}\,t^3.$$

Applied in an IVP. Solve $y''+2y'+y = te^{-t}$, $y(0)=y'(0)=0$ (forcing at the double root -1). The forcing transforms by the s-shift too: $\mathcal{L}\{te^{-t}\} = \frac{1}{(s+1)^2}$, so $(s+1)^2Y = \frac{1}{(s+1)^2}$, giving $Y = \frac{1}{(s+1)^4}$ and $y(t) = \frac{1}{6}t^3e^{-t}$.

Completing the square

When a quadratic denominator has *complex* roots — discriminant negative, so it will not factor into real linear pieces — it cannot be split by ordinary partial fractions. Instead we reshape it into a perfect square plus a constant and read off a damped sine/cosine. The two forms this produces are the ones to memorise:

The two workhorse forms

$$\mathcal{L}^{-1}\left\{\frac{1}{(s-a)^2+b^2}\right\} = \frac{1}{b}e^{at}\sin bt, \qquad \mathcal{L}^{-1}\left\{\frac{s-a}{(s-a)^2+b^2}\right\} = e^{at}\cos bt.$$

The recipe for $\dfrac{\alpha s+\beta}{s^2+ps+q}$ (complex roots, $p^2<4q$)

1. **Complete the square:** $s^2+ps+q=(s-a)^2+b^2$, with $a=-\dfrac{p}{2}$, $b=\sqrt{q-\dfrac{p^2}{4}}$.
2. **Match the numerator to $s-a$:** $\alpha s+\beta=\alpha\,(s-a)+(\beta+\alpha a)$.
3. **Split and invert** using the two forms above:

$$\mathcal{L}^{-1}\left\{\frac{\alpha s+\beta}{s^2+ps+q}\right\}=\alpha\,\mathrm{e}^{at}\cos bt+\frac{\beta+\alpha a}{b}\,\mathrm{e}^{at}\sin bt.$$

The whole trick is Step 2: force the numerator to show the same $s-a$ that sits inside the square, so a clean cosine peels off and whatever is left over is a sine.

Example 1 — pure sine

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2+2s+5}\right\}.$$

Complete the square: $s^2+2s+5=(s+1)^2+4$, so $a=-1$, $b=2$. The numerator is a bare constant, so only a sine appears:

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+1)^2+2^2}\right\}=\frac{1}{2}\,\mathrm{e}^{-t}\sin 2t.$$

Applied in an IVP. This is the response of an under-damped oscillator released with unit velocity: $y''+2y'+5y=0$, $y(0)=0$, $y'(0)=1$ gives $(s^2+2s+5)Y=1$, so $Y=\frac{1}{s^2+2s+5}$ and $y(t)=\frac{1}{2}\mathrm{e}^{-t}\sin 2t$.

Example 2 — cosine and sine together

$$\mathcal{L}^{-1}\left\{\frac{s+1}{s^2+4s+13}\right\}.$$

Square: $s^2+4s+13=(s+2)^2+9$, so $a=-2$, $b=3$. Match the numerator: $s+1=(s+2)-1$, hence

$$\frac{s+1}{(s+2)^2+9}=\frac{s+2}{(s+2)^2+9}-\frac{1}{(s+2)^2+9},$$

which inverts term by term to

$$\mathrm{e}^{-2t}\cos 3t-\tfrac{1}{3}\mathrm{e}^{-2t}\sin 3t=\mathrm{e}^{-2t}\left(\cos 3t-\tfrac{1}{3}\sin 3t\right).$$

Applied in an IVP. Exactly this $Y(s)$ arises from $y''+4y'+13y=0$, $y(0)=1$, $y'(0)=-3$: then $(s^2+4s+13)Y=s+1$, so $Y=\frac{s+1}{s^2+4s+13}$ and $y(t)=\mathrm{e}^{-2t}\left(\cos 3t-\frac{1}{3}\sin 3t\right)$.

Example 3 — a general numerator

$$\mathcal{L}^{-1}\left\{\frac{2s-3}{s^2-4s+8}\right\}.$$

Square: $s^2-4s+8=(s-2)^2+4$, so $a=2$, $b=2$. Match: $2s-3=2(s-2)+1$, then

$$\frac{2s-3}{(s-2)^2+4}=2\cdot\frac{s-2}{(s-2)^2+4}+\frac{1}{(s-2)^2+4},$$

which inverts term by term to

$$2\mathrm{e}^{2t}\cos 2t+\tfrac{1}{2}\mathrm{e}^{2t}\sin 2t=\mathrm{e}^{2t}\left(2\cos 2t+\tfrac{1}{2}\sin 2t\right).$$

Applied in an IVP. Here the roots $2 \pm 2i$ have positive real part — a growing oscillation. Solve $y'' - 4y' + 8y = 0$, $y(0) = 2$, $y'(0) = 5$: $(s^2 - 4s + 8)Y = 2s - 3$, so $Y = \frac{2s-3}{s^2-4s+8}$ and $y(t) = \mathrm{e}^{2t}\big(2\cos 2t + \frac{1}{2}\sin 2t\big)$.

Multiplication by t

Differentiation of the transform

$$\mathcal{L}\{t\,f(t)\} = -F'(s), \qquad \text{and more generally} \qquad \mathcal{L}\{t^n f(t)\} = (-1)^n F^{(n)}(s).$$

Why it holds. Differentiate the defining integral under the integral sign:

$$F'(s) = \frac{\mathrm{d}}{\mathrm{d}s}\int_0^\infty \mathrm{e}^{-st} f(t)\,\mathrm{d}t = \int_0^\infty (-t)\,\mathrm{e}^{-st} f(t)\,\mathrm{d}t = -\,\mathcal{L}\{t\,f(t)\}.$$

Each extra power of t costs one more derivative in s (and one more sign).

Where it earns its keep. Applied to a sine or cosine it produces the *repeated* quadratic denominators that ordinary partial fractions cannot reach — exactly the shapes that appear in **resonance**:

Two transforms worth knowing by heart

$$\mathcal{L}\{t\sin\omega t\} = \frac{2\omega s}{(s^2+\omega^2)^2}, \qquad \mathcal{L}\{t\cos\omega t\} = \frac{s^2-\omega^2}{(s^2+\omega^2)^2}.$$

Example 1 — deriving $\mathcal{L}\{t\sin\omega t\}$

Start from $F(s) = \mathcal{L}\{\sin\omega t\} = \dfrac{\omega}{s^2+\omega^2}$. Differentiate:

$$F'(s) = \omega \cdot \frac{-2s}{(s^2+\omega^2)^2} \implies \mathcal{L}\{t\sin\omega t\} = -F'(s) = \frac{2\omega s}{(s^2+\omega^2)^2}.$$

Applied in an IVP. With $\omega = 1$ this inverts the resonant response to a cosine drive at the natural frequency: $y'' + y = 2\cos t$, $y(0) = y'(0) = 0$ gives $Y = \frac{2s}{(s^2+1)^2}$, so $y(t) = t\sin t$ — an amplitude growing linearly in t.

Example 2 — deriving $\mathcal{L}\{t\cos\omega t\}$

Now $F(s) = \mathcal{L}\{\cos\omega t\} = \dfrac{s}{s^2+\omega^2}$. By the quotient rule,

$$F'(s) = \frac{(s^2+\omega^2) - s(2s)}{(s^2+\omega^2)^2} = \frac{\omega^2 - s^2}{(s^2+\omega^2)^2} \implies \mathcal{L}\{t\cos\omega t\} = -F'(s) = \frac{s^2-\omega^2}{(s^2+\omega^2)^2}.$$

Applied in an IVP. The $t\cos\omega t$ term is the hallmark of a *sine*-driven resonance. Solve $y'' + y = \sin t$, $y(0) = y'(0) = 0$: then $Y = \frac{1}{(s^2+1)^2} = \frac{1}{2}\left[\frac{1}{s^2+1} - \frac{s^2-1}{(s^2+1)^2}\right]$, and inverting the second piece by this example gives $y(t) = \frac{1}{2}(\sin t - t\cos t)$.

Example 3 — inverting a repeated quadratic

$$\mathcal{L}^{-1}\left\{\frac{s}{(s^2+9)^2}\right\}.$$

This matches the $\mathcal{L}\{t\sin\omega t\}$ shape with $\omega = 3$, for which $\dfrac{2\omega s}{(s^2+\omega^2)^2} = \dfrac{6s}{(s^2+9)^2}$. Our expression is one-sixth of that, so

$$\mathcal{L}^{-1}\left\{\frac{s}{(s^2+9)^2}\right\} = \frac{1}{6}\,t\sin 3t.$$

This is precisely the kind of term that shows the unbounded growth of a resonant response.

Applied in an IVP. It is the response to a cosine drive at the natural frequency 3: $y''+9y = \cos 3t$, $y(0)=y'(0)=0$ gives $(s^2+9)Y = \frac{s}{s^2+9}$, so $Y = \frac{s}{(s^2+9)^2}$ and $y(t) = \frac{1}{6}\,t\sin 3t$.

6. Solving initial-value problems: a gallery

Every problem below runs the same four beats — *transform* (the derivative rule folds in the data at once), *collect* $Y(s)$, *split* by partial fractions, *invert*. Watch how the **form of the forcing** and the **roots of the denominator** together shape the answer.

A — forced, distinct real roots

Solve $y''-3y'+2y = 4, \qquad y(0)=1,\ y'(0)=0.$

Transform every term, using the derivative rule and $\mathcal{L}\{4\} = 4/s$, then insert the data:

$$\underbrace{\big[s^2Y - s\,y(0) - y'(0)\big]}_{\mathcal{L}\{y''\}} - 3\underbrace{\big[sY - y(0)\big]}_{\mathcal{L}\{y'\}} + 2Y = \frac{4}{s} \implies (s-1)(s-2)\,Y = \frac{s^2-3s+4}{s}.$$

Partial fractions, then invert ($\frac{1}{s}\to 1$, $\frac{1}{s-a}\to \mathrm{e}^{at}$):

$$Y = \frac{2}{s} - \frac{2}{s-1} + \frac{1}{s-2} \implies \boxed{y(t) = 2 - 2\mathrm{e}^{t} + \mathrm{e}^{2t}.}$$

Check: $y(0)=1$, $y'(0)=0$ — satisfied by the algebra, not fitted afterwards.

Contrast with Unit 2. There we would find the complementary function $c_1\mathrm{e}^{t} + c_2\mathrm{e}^{2t}$, then a particular integral, then solve two simultaneous equations for c_1, c_2. Laplace collapses all of that into one algebraic solve, and the answer is already the specific solution to the IVP.

B — sinusoidal forcing (non-resonant)

Solve $y''+4y = \cos t, \qquad y(0)=0,\ y'(0)=0.$

With $\mathcal{L}\{\cos t\} = \frac{s}{s^2+1}$,

$$(s^2+4)\,Y = \frac{s}{s^2+1} \implies Y = \frac{s}{(s^2+1)(s^2+4)} = \frac{1}{3}\,\frac{s}{s^2+1} - \frac{1}{3}\,\frac{s}{s^2+4}.$$

Inverting each cosine term,

$$\boxed{y(t) = \tfrac{1}{3}\big(\cos t - \cos 2t\big).}$$

Both the forcing frequency (1) and the natural frequency (2) appear in the response.

C — complex roots: a damped oscillation

Solve $y'' + 4y' + 13y = 0, \qquad y(0) = 1,\ y'(0) = -2.$

Transform with the data ($\mathcal{L}\{y''\} = s^2Y - s + 2$, since $y'(0) = -2$):

$$\big[s^2Y - s + 2\big] + 4\big[sY - 1\big] + 13Y = 0 \implies (s^2 + 4s + 13)\,Y = s + 2.$$

Complete the square, $s^2 + 4s + 13 = (s+2)^2 + 9$, matching the $\frac{s-a}{(s-a)^2+b^2}$ form:

$$Y = \frac{s+2}{(s+2)^2 + 9} \implies \boxed{y(t) = \mathrm{e}^{-2t}\cos 3t.}$$

The complex roots $-2 \pm 3i$ read straight off as decay rate 2 and ringing frequency 3.

D — forcing plus non-zero data: free + forced response

Solve $y'' + y = \sin 2t, \qquad y(0) = 1,\ y'(0) = 0.$

With $\mathcal{L}\{\sin 2t\} = \frac{2}{s^2+4}$,

$$(s^2+1)\,Y = s + \frac{2}{s^2+4} \implies Y = \underbrace{\frac{s}{s^2+1}}_{\text{from the data}} + \underbrace{\frac{2}{(s^2+1)(s^2+4)}}_{\text{from the forcing}}.$$

The second piece is $\frac{2}{3}\frac{1}{s^2+1} - \frac{2}{3}\frac{1}{s^2+4}$, so inverting all terms,

$$\boxed{y(t) = \cos t + \tfrac{2}{3}\sin t - \tfrac{1}{3}\sin 2t.}$$

The $\cos t$ is the *free vibration* carried by the initial data; the rest is the response to the drive — superposition, visible term by term.

E — resonance: forcing at the natural frequency

Solve $y'' + y = \cos t, \qquad y(0) = 0,\ y'(0) = 0.$

Now the drive sits *exactly* at the natural frequency, and the two s^2+1 factors collide:

$$(s^2+1)\,Y = \frac{s}{s^2+1} \implies Y = \frac{s}{(s^2+1)^2}.$$

This is the repeated-quadratic form from the multiplication-by-t rule (with $\omega = 1$):

$$\boxed{y(t) = \tfrac{t}{2}\sin t.}$$

The amplitude grows without bound — Laplace produces the resonant (*secular*) term automatically, with no special case to remember.

F — first order: transient meets steady state

Solve $y' + y = \cos t, \qquad y(0) = 0.$

Transforming, $(s+1)Y = \frac{s}{s^2+1}$, so

$$Y = \frac{s}{(s+1)(s^2+1)} = -\frac{1}{2}\frac{1}{s+1} + \frac{1}{2}\frac{s+1}{s^2+1} \implies \boxed{y(t) = \tfrac{1}{2}\big(\cos t + \sin t - \mathrm{e}^{-t}\big).}$$

The e^{-t} is a *transient* that dies away; what remains is the sinusoidal *steady state* the system settles into.

7. . . . and more: what Laplace does that the earlier methods cannot

Every method so far — undetermined coefficients, inverse operators, even the gallery above — quietly assumes the forcing $f(t)$ is *smooth and built from the usual pieces*: exponentials, polynomials, sines and cosines. Real inputs are often not so tidy. A switch is flipped at a set time (a *discontinuous* input); a hammer strikes the system (an *impulsive* input); or the drive is an arbitrary recorded signal with no formula at all. Laplace handles all three, because in the s-domain each becomes a simple algebraic factor. This section introduces the three constructs that make that possible — the *unit step*, the *Dirac delta*, and *convolution*.

The unit step: switching forcing on and off

Unit step (Heaviside) function

$$u(t-a)=\begin{cases}0, & t<a,\\ 1, & t\geq a,\end{cases}\qquad \mathcal{L}\{u(t-a)\}=\frac{\mathrm{e}^{-as}}{s}\quad (a\geq 0).$$

Think of $u(t-a)$ as a master switch that turns on at $t=a$. Multiplying a term by it delays that term's arrival to $t=a$; subtracting two steps makes a *window*,

$$u(t-a)-u(t-b)=\begin{cases}1, & a\leq t<b,\\ 0, & \text{otherwise},\end{cases}$$

on only between a and b. With these, *any* piecewise-defined forcing becomes a single formula — for instance

$$f(t)=\begin{cases}2, & 0\leq t<3,\\ 5, & t\geq 3,\end{cases}\qquad\Longleftrightarrow\qquad f(t)=2+3\,u(t-3).$$

To transform such things we need the companion of the s-shift:

Second shifting theorem (the t-shift)

$$\mathcal{L}\{u(t-a)\,f(t-a)\}=\mathrm{e}^{-as}F(s).$$

A signal held off until $t=a$ and then run *with its clock reset to* a picks up the factor e^{-as}.

Mind the argument. The theorem needs the *same* shift $t-a$ inside the function. If you meet $u(t-a)\,g(t)$ with g not already written in $(t-a)$, first re-express g as a function of $t-a$ — e.g. $t^2=\big((t-2)+2\big)^2=(t-2)^2+4(t-2)+4$.

Transforming a delayed term by the rewrite trick

$\mathcal{L}\{u(t-2)\,t^2\}$.

Rewrite t^2 around $t=2$: $t^2=(t-2)^2+4(t-2)+4$. Now every piece carries the shift $t-2$, so the second shift applies term by term:

$$\mathcal{L}\{u(t-2)\,t^2\}=\mathrm{e}^{-2s}\left(\frac{2}{s^3}+\frac{4}{s^2}+\frac{4}{s}\right).$$

A system kicked into life at $t = \pi$

Solve $y'' + y = u(t-\pi)$, $y(0) = 0$, $y'(0) = 0$ — an oscillator at rest, then forced from $t = \pi$ on.

Transforming, $(s^2+1)Y = \dfrac{\mathrm{e}^{-\pi s}}{s}$, so $Y = \mathrm{e}^{-\pi s}\cdot\dfrac{1}{s(s^2+1)} = \mathrm{e}^{-\pi s}\left(\dfrac{1}{s} - \dfrac{s}{s^2+1}\right)$. Since $\mathcal{L}^{-1}\left\{\frac{1}{s(s^2+1)}\right\} = 1-\cos t$, the second shift attaches $u(t-\pi)$ and delays the argument by π:

$$y(t) = u(t-\pi)\big[\,1-\cos(t-\pi)\,\big] = u(t-\pi)\big(1+\cos t\big).$$

The mass sits still until $t = \pi$, then responds — a genuinely piecewise answer, produced without ever splitting the problem into cases by hand.

The Dirac delta: an instantaneous impulse

A hammer blow, a voltage spike, a bat meeting a ball — a large force acting over a negligibly short time, delivering a definite total *impulse*. Idealising "short" to instantaneous gives the *Dirac delta* $\delta(t-a)$: zero everywhere except at $t = a$, with unit area concentrated there. Its defining feature is the **sifting property**, which reads off the value of any continuous g at the spike:

$$\int_0^\infty \delta(t-a)\,g(t)\,\mathrm{d}t = g(a).$$

Transform of an impulse

Taking $g(t) = \mathrm{e}^{-st}$ in the sifting property gives it at once:

$$\mathcal{L}\{\delta(t-a)\} = \mathrm{e}^{-as}, \qquad \text{and in particular} \qquad \mathcal{L}\{\delta(t)\} = 1.$$

Feed a unit impulse into a system at rest and the output is its *impulse response*. For $y'' + y = \delta(t)$, $y(0) = y'(0) = 0$:

$$(s^2+1)Y = 1 \implies Y = \frac{1}{s^2+1} \implies h(t) = \sin t.$$

The factor $H(s) = \dfrac{1}{s^2+1}$ is the **transfer function**: it multiplies whatever you put in, $Y(s) = H(s)\cdot(\text{input})$, and depends only on the system, not on the forcing. The impulse response is simply its inverse transform, $h = \mathcal{L}^{-1}\{H\}$.

A delayed blow

Solve $y'' + y = \delta(t-a)$, $y(0) = 0$, $y'(0) = 0$.

Transforming, $(s^2+1)Y = \mathrm{e}^{-as}$, so $Y = \mathrm{e}^{-as}\cdot\dfrac{1}{s^2+1}$, and the second shift gives

$$y(t) = u(t-a)\,\sin(t-a).$$

The oscillator is perfectly still until the blow lands at $t = a$, then rings sinusoidally ever after.

Convolution: the response to *any* forcing

When a transform factors as a *product*, its inverse is not the product of the inverses but their **convolution**:

Convolution theorem

$$(f * g)(t) = \int_0^t f(\tau)\, g(t-\tau)\, \mathrm{d}\tau, \qquad \mathcal{L}\{f * g\} = F(s)\, G(s).$$

The operation is commutative, $f * g = g * f$.

Inverting a product

$\mathcal{L}^{-1}\left\{\dfrac{1}{s^2(s^2+1)}\right\}$.

Read the denominator as $\dfrac{1}{s^2} \cdot \dfrac{1}{s^2+1}$, i.e. $F(s)G(s)$ with $f(t) = t$ and $g(t) = \sin t$. Then

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2(s^2+1)}\right\} = (t * \sin t) = \int_0^t \tau \, \sin(t-\tau)\, \mathrm{d}\tau = t - \sin t.$$

The real prize is a solution formula for *arbitrary* forcing. Combining the last two ideas: the transfer function turns the transformed equation into the product $Y(s) = H(s)F(s)$, so in the time domain the solution is the impulse response convolved with the input,

$$\boxed{y(t) = (h * f)(t) = \int_0^t h(t-\tau)\, f(\tau)\, \mathrm{d}\tau.}$$

This is *Duhamel's principle*: find the system's impulse response once, and the response to *any* input is a single convolution — no fresh particular integral for each new f.

One impulse response, any input

For $y'' + y = f(t)$ at rest the impulse response is $h(t) = \sin t$, so $y(t) = \displaystyle\int_0^t \sin(t-\tau)\, f(\tau)\, \mathrm{d}\tau$ for *every* f. Taking $f(t) = \mathrm{e}^t$, for example,

$$y(t) = \int_0^t \sin(t-\tau)\, \mathrm{e}^{\tau}\, \mathrm{d}\tau = \tfrac{1}{2}\big(\mathrm{e}^t - \cos t - \sin t\big).$$

Looking ahead. The pair "transfer function $H(s)$ + input" is the working language of control engineering and signal processing: a system's stability, resonances, and response to any signal are read directly from $H(s)$ in the s-domain, and the impulse response $h(t)$ is its time-domain fingerprint. What begins here as a device for initial-value problems is, in that sense, the foundation of how dynamic systems are modelled and controlled.

8. Reference: a working table of transforms

$f(t) \quad \longmapsto \quad F(s) = \mathcal{L}\{f(t)\}$

$f(t)$	$F(s)$
1	$\dfrac{1}{s}$
t^n	$\dfrac{n!}{s^{n+1}}$
e^{at}	$\dfrac{1}{s-a}$
$\sin \omega t$	$\dfrac{\omega}{s^2+\omega^2}$
$\cos \omega t$	$\dfrac{s}{s^2+\omega^2}$
$\mathrm{e}^{at}\sin bt$	$\dfrac{b}{(s-a)^2+b^2}$
$\mathrm{e}^{at}\cos bt$	$\dfrac{s-a}{(s-a)^2+b^2}$
$u(t-a)$	$\dfrac{\mathrm{e}^{-as}}{s}$
$\delta(t-a)$	e^{-as}
$f'(t)$	$sF(s)-f(0)$
$f''(t)$	$s^2F(s)-sf(0)-f'(0)$
$\mathrm{e}^{at}f(t)$	$F(s-a)$ (first shift)
$u(t-a)\,f(t-a)$	$\mathrm{e}^{-as}F(s)$ (second shift)
$(f*g)(t)$	$F(s)\,G(s)$ (convolution)

9. Practice

Answers are given so you can check yourself; the method matters more than the final line.

Exercises

1. Transform $3 + 2t - \mathrm{e}^{-t} + \sin 2t$.

2. Invert $\dfrac{s+3}{s^2+4s+13}$.

3. Solve $y' + 3y = \mathrm{e}^{-3t}, \quad y(0) = 1$.

4. Solve $y'' + 4y = 0, \quad y(0) = 0,\ y'(0) = 2$.

5. Solve $y'' - y = u(t-1), \quad y(0) = 0,\ y'(0) = 0$.

Answers. **1.** $\dfrac{3}{s} + \dfrac{2}{s^2} - \dfrac{1}{s+1} + \dfrac{2}{s^2+4}$. **2.** $\mathrm{e}^{-2t}\big(\cos 3t + \tfrac{1}{3}\sin 3t\big)$. **3.** $y = (1+t)\mathrm{e}^{-3t}$. **4.** $y = \sin 2t$. **5.** $y = u(t-1)\big[\cosh(t-1) - 1\big]$.

End of Unit 3 introduction — M3503, Department of Mathematics & Computer Science, NUL.